Supplement: methods, sensitivities and audit trail

For "Older than their fossils? What the Ordovician–Silurian record can test about the maxima that divide land-plant molecular clocks". Internal research draft; not for release.

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Table of contents

S1. Readings, observations and units . . . 1
S2. The Murzuq encounter comparison . . . 2
S2.1 Source and frame . . . 2
S2.2 Model and qualification . . . 2
S2.3 Contamination and an identification limit . . . 3
S3. QSIM-801: a model-conditional floor calculation . . . 5
S4. The Cambrian affinity review (M1) . . . 5
S5. The positive body-record analysis . . . 7
S5.1 Inventory and units . . . 7
S5.2 The endpoint calculation . . . 7
S5.3 Results and sensitivities . . . 8
S5.4 Recovery before the oldest record . . . 12
S5.5 Biological delay . . . 13
S5.6 Simulation checks . . . 13
S5.7 Figures from the reviewed analysis . . . 17
S6. From specimens to nodes . . . 17
S7. Analyses superseded during review . . . 17
S8. Reproduction and navigation . . . 20
Supplement references . . . 20

This supplement holds the machinery behind the main text. Repository paths are given so that every number can be traced to the table or script that produced it. Internal labels appear here because the repository uses them: **M1**, **M2** and **M3** are the 515.5, 459.1 and 431.2 Ma maxima; **E1** trilete spores, **E2** cryptospores accepted as embryophytic under a named reading, **E3** polysporangiate bodies, and **E5** crown vascular-plant members. **B0**, **B1** and **B2** are the body-record categories: any land-plant body, polysporangiophyte body and crown vascular-plant member.

S1. Readings, observations and units

Named readings. Disputed identifications, affinities and ages are carried as alternatives, not resolved by fiat. The *main* reading accepts what the cited sources accept. The *inclusive* reading adds records that a named source accepts against others, and the *outlier* reading admits further records that are

explicitly contested. A wider reading requires a record-specific acceptance; it does not promote every unresolved grain or axis to a detection. The E2 rule separates spore morphology and producer evidence from the embryophyte affinity accepted by each reading (`data/units/e2-unit-rule.md`). Rejected, reassigned and disputed records are listed in `docs/rejected-records.md`.

Observations. Within each frame, the same reporting rule applies to older and reference material. An explicit non-detection is scoped to the material actually examined. A missing report becomes a zero only after a full-category reporting audit. Unresolved identification, missing links and unreported effort remain distinct from absence.

Ages. Stage bases follow ICS v2024/12 (`data/timescale.csv`). Source-specific age statements and their numerical conversions are retained separately. Literature before 2024 mostly uses GTS2012 Early Devonian boundaries, which differ from ours by up to 3 Myr. Uniform within-range ages and shared within-range positions are modelling choices, not measured dating distributions.

S2. The Murzuq encounter comparison

S2.1 Source and frame

Abuhmida (2013) reports palynology from six Murzuq Basin wells. All 164 appendix samples were transcribed independently, with 158 chart depth and type labels (153 exact matches). The 164 prose-scope rows and 158 chart reports were mapped, retaining barren material and possible caving (`data/units/libya_*.csv`; `docs/audits/2026-10-31-libya-event-map.md`). The primary frame is the chart-coded cuttings design `LY-MZ-CU`. Five wells contribute 59 Llandovery reference cuttings (51 with trilete reports), and three of them contribute 28 older cuttings without trilete reports.

S2.2 Model and qualification

For an eligible unit, the detection probability conditional on a well effect is $1 - \exp(-rux)$. Here r is the reference rate, $u \sim \text{Gamma}(2, 2)$ is shared within a well, and $x = \rho f$ is the older relative intensity. The older/reference delivery-and-recovery ratio ρ takes the declared values 1, 0.5, 0.25 and 0.1; it is not estimated from the source. Reference and older outcomes share the well effect and are fitted jointly.

H0c constrains constant intensity; H0 and H0m constrain exposure-weighted intensity under their declared history classes; H1 allows the declared higher ceiling. The origin read-out at 459.1 Ma concerns the interval from the reference boundary $\tau = 443.1$ Ma to that age. A window comparison instead tests a persistent minimum intensity on an older interval beginning at 459.1 Ma, allowing zero intensity between that boundary and τ. An unbounded low-intensity window therefore does not show that the same intensity is compatible all the way to the reference.

Inference is by test inversion with a declared, simulation-calibrated critical value (`docs/f3-shape.md`). The potential screen, eligibility and estimation path were simulated on a declared finite grid. Results are nominal 95% compatibility bounds, qualified per quantity and design against a 0.93 coverage floor (calibration target 0.955), conditional on passing the potential screen, with abstentions counted as covering. The largest observed marginal miss proportion was 38/1,000. At the B4 boundary of the 462.6 Ma window, 20 of 1,000 screen-passing simulations excluded the truth, and 60 delivered non-abstaining window claims, so 20 of those 60 missed. Population 95% coverage, coverage conditional on a non-abstaining result, simultaneous coverage and validity beyond the tested histories

are not established. Thirteen Murzuq design/delivery combinations qualify. A fourteenth qualified pair, QB-CO, supplies no older M2 interval and has a narrower class scope. The receipts are in `models/f3/out/qual/`.

S2.3 Contamination and an identification limit

Murzuq's palynofacies fractions use a different preparation and a 200-particle denominator. They cannot supply separate trilete and cryptospore counts for the chart search, so binary observations are retained and contamination enters as a sensitivity model. Let g be the indigenous fraction of reference *encounter intensity* (not of positive samples), and κ the transfer of the contaminant component into older cuttings. For age-exposure weight q_i, an older zero has probability

$$\exp[-ru\kappa(1-g)]\{(1-q_i)+q_i\exp[-ru\rho gf]\}.$$

The likelihood depends on these parameters only through $a = \rho gf$ and $b = \kappa(1-g)$, so tuples preserving both are observationally equivalent within the model. If every $q_i = 1$, only their sum enters the older-zero likelihood. Declining indigenous output can thus be offset by delivery and reference contamination along stated parameter combinations. This is an identification result within the declared model, not a universal property of binary observations.

A predeclared 5,760-row factorial grid varies delivery, indigenous fraction, transfer, allocation, chronology, interval and intensity. Independent uniform ages are compared with fixed, depth-ordered 10%, 50% and 90% positions within source ranges. The threshold $-2\log(0.05)$ is an uncalibrated comparison convention. A contamination scenario that conflicts with older zeros even at $f = 0$ is flagged separately. The 384 break-even entries fix $\kappa = b = 0$. These surfaces and the 36-case well/age influence exercise are unqualified sensitivities.

Table S 1: Murzuq sensitivities (`models/f3/loop2/out/`; `docs/loop2-sprint1-results.md`).

Challenge at $\rho = 1$	Observed sensitivity	Scope
Five leave-one-well-out cases	H0 0.1850–0.3646; window endpoints 462.6–468.1 Ma	one factor at a time; unqualified
All nine influence/age/allocation cases	H0 0.1775–0.3646	unqualified
$\kappa = 0$, independent uniform stage ages	critical g: 0.0910 for constant origin intensity; 0.9559, 0.2363, 0.1735 for windows ending 462.6, 467.1, 469.4 Ma	$f = 1$; comparison contours, not confidence limits
Shared midpoint ages	no exposure in the 459.1–462.6 Ma window	fixed chronology scenario
Shared 90% ages	no exposure in the window ending at 467.1 Ma	fixed chronology scenario

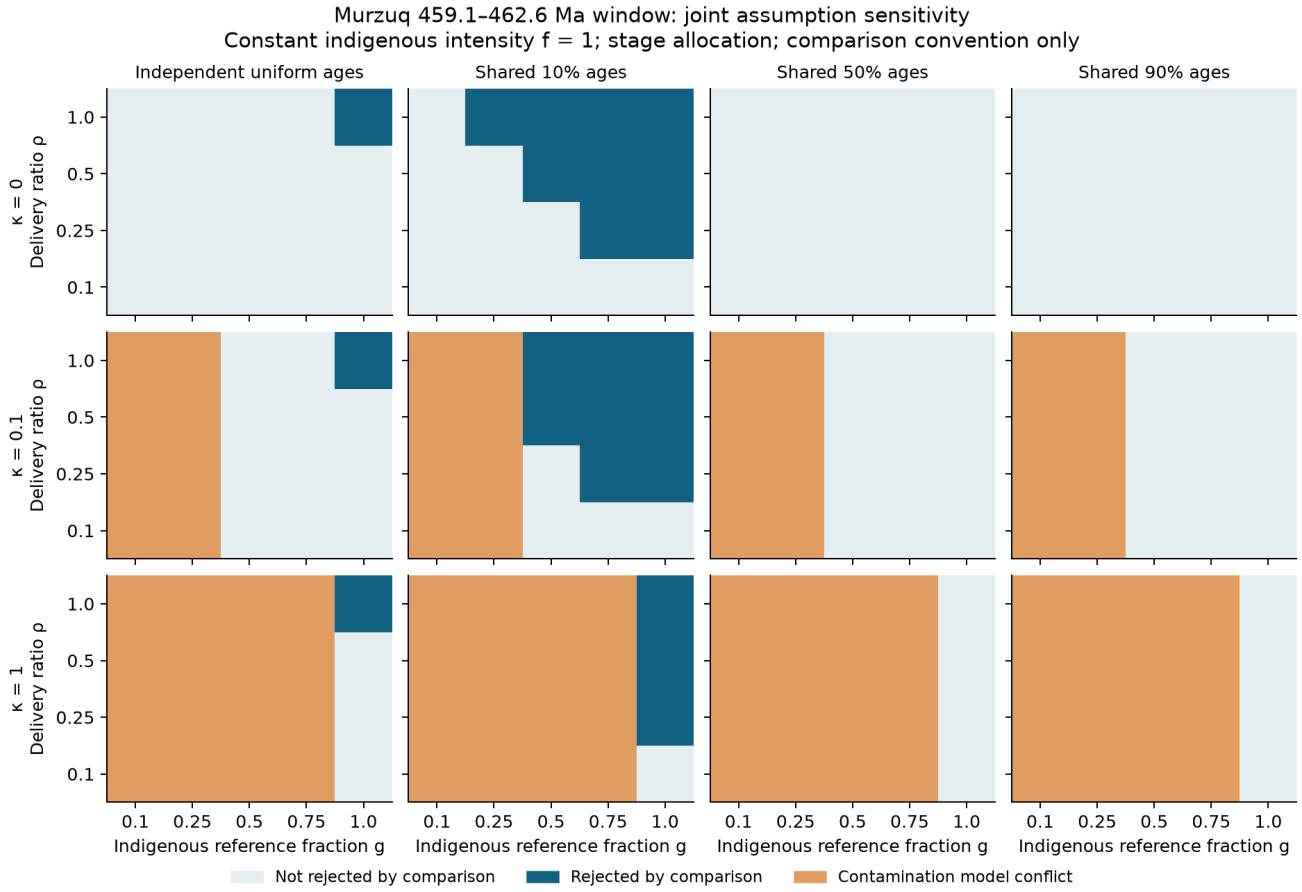


Figure S 1: Joint sensitivity of the short Murzuq window. Colours report an uncalibrated comparison or model-conflict flag, not confidence or probabilities over geology.

S3. QSIM-801: a model-conditional floor calculation

Vecoli et al. (2017, Table 1, p. 321) report 99 cryptospore morphotaxon occurrences across 17 of 20 tabled samples assigned to the late Dapingian–early Darriwilian. Their p. 330 explicitly reports no triletes in the studied sequence. The table was independently reconstructed from the rendered source (`data/units/loop2/replication/`). Its 22 categories include informal morphotypes. Marks are distinct recorded occurrences, not counts of independent producers or dispersal events. The table-defined depths are 675.1–767.8 ft. The methods' shallower endpoint, 586.9 ft, is inconsistent and is not used.

Let θ_i be the trilete/control unit-intensity ratio in sample i, and $h > 0$ a relative effective recovery factor. Suppose the reported objects form a co-ascertained, independent target/control sample. Conditioning Poisson counts on the true land-spore total N_i then gives target probability $h\theta_i/(1 + h\theta_i)$. For a persistent floor $\psi = \min_i \theta_i \geq \theta$, the all-zero probability is at most $(1 + h\theta)^{-\sum_i N_i}$. If each mark contributes a distinct target-detectable unit, $M \leq \sum_i N_i$ permits a conservative rejection rule: no targets and $(1+h\theta)^{-M} \leq \alpha$. Randomness of M does not invalidate the rule: conditional on the actual totals, rejection is a subset of the all-zero event with enough units to cross the threshold. Then $U = \{\alpha^{-1/M} - 1\}/h$ bounds the *minimum* ratio, not every sample's ratio, a mean, or the pooled share.

Table S 2: Model-conditional floor limits; none is an admitted empirical confidence bound. D divides the exponent as a heuristic for clustering, not a validated correction.

Relative effective recovery h	Independent units, $D = 1$	Heuristic $D = 2$	Heuristic $D = 4$
1	0.031	0.062	0.129
0.5	0.061	0.125	0.257
0.25	0.123	0.250	0.515
0.1	0.307	0.624	1.287

The source specifies preparation but not the search coverage needed to show that every mark represents a unit on which triletes were detectable. Independence, effective ascertainment and sampling and stopping conditions are unverified. The calculation therefore stays exploratory. Toy Poisson simulations check the algebra but cannot validate the historical protocol. No matched counted Katian–Llandovery reference was recovered. Qusaiba-1 supplies reference context only. Röstånga and Caspalá/Lipeón are younger than M2, and Hanadir/TAYM-4 is separate material without a qualified design (`manuscript/external_candidates.csv`).

S4. The Cambrian affinity review (M1)

No assessed frame admits an E2 older-zero/reference design. The Borenshult 71.1 m zero is secure under the main reading but belongs to a frame that fails its potential screen. It is not a pre-515.5 Ma observation. Strother (2016) places the Tennessee taxa in Charophyta. Taylor & Strother (2024) interpret the packets as pre-embryophytic, and Wang et al. (2022) favour ancestral aeroterrestrial algae while calling direct embryophyte affinity unlikely and the interpretation premature. None explicitly accepts the assessed Stage 4 producers as embryophytes. Wall ultrastructure and developmental

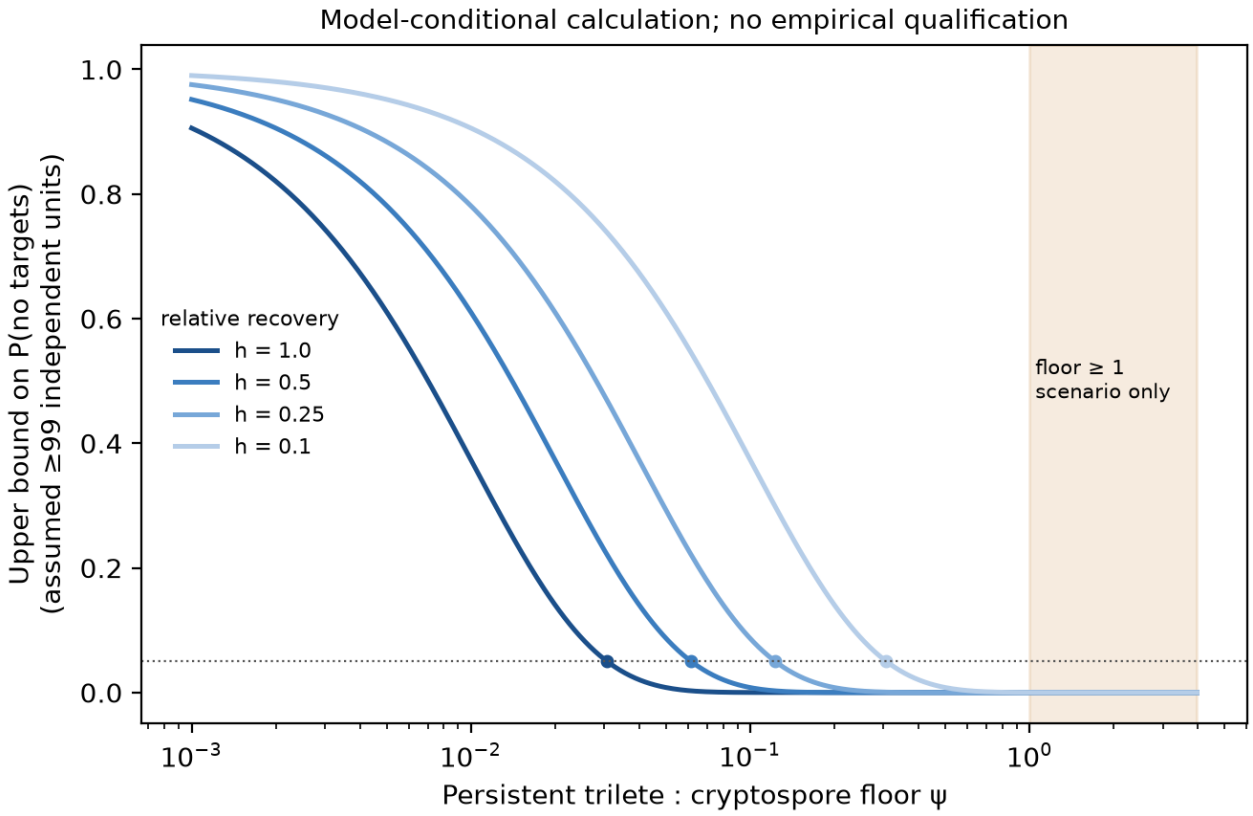


Figure S 2: Upper bound on the all-zero probability under the hypothetical independent, co-ascertained unit model; points mark the 0.05 threshold.

features bear on precursor relationships, so the inclusive reading stays an explicit alternative. Argentine sample RIOCAP-7999 is unresolved under main and detected under inclusive and outlier readings. Stage 4 converts to 514.5–506.5 Ma, all younger than 515.5 Ma (`data/units/loop2/m1-m3/`).

S5. The positive body-record analysis

S5.1 Inventory and units

Names. The 241 distinct PBDB macrofossil names overlapping 445.2–393.47 Ma were classified by `gpt-6.1-sol` with web search. Each name received a category, polysporangiophyte and crown-member flags, a basis, a cited source and a confidence (`data/macro/name_classifications.csv`). Names judged implausible for the window, for example the Cretaceous angiosperm leaf genus *Protophyllum*, cannot enter any category. A stratified source audit is in `docs/audits/2026-11-01-loop3-source-review` it does not certify all 241 assignments.

Material identity. Only explicit collection or occurrence identifiers justify replacing a PBDB record with its curated interpretation. Country, taxon and overlapping ages do not establish identity; the pilot used them and falsely merged distinct localities (S7). Regional checklists and unresolved duplicates are withheld from the primary count and kept in a sensitivity. Source decisions are in `data/macro/source_adjudications.csv`. Example: the two Yea horizons are kept as separate units with the full disputed Gorstian–Emsian range, because the Gorstian attribution rests on a disputed graptolite identification. Neither a fixed Gorstian nor a fixed Devonian age is justified.

Units. The primary unit is a locality–age proxy: PBDB formation plus coordinates rounded to 0.1° and identical age support, or a curated named site plus identical age support. For a collapsed unit, the oldest-member age has support from the maximum of member lower ages to the maximum of member upper ages. Missing geographic keys retain the collection identifier. The proxy is conservative but is not a verified bed (`models/f3/loop3/out/primary_unit_membership.csv` lists every constituent record).

Older screen. A separate screen covers all 556 older-only non-spore PBDB rows (69 names). Source review of the seven plausible or unplaced names (17 occurrences) found cyanobacteria, algae or trace fossils and the rejected *Boiophyton*, not an additional embryophyte body. These exclusions are not fossil non-detections, and the screen is not a search denominator (`data/macro/older_records_screen.csv`; `docs/audits/2026-11-01-loop3-older-records-review.md`).

S5.2 The endpoint calculation

For a fixed younger cutoff c, let $U(a) = \int_c^a w(s)\,ds$ be cumulative recovery. Conditional on n independent units with ages drawn with density proportional to w on $[c, T]$, the maximum A satisfies $P(A \le a) = \{U(a)/U(T)\}^n$. Inverting at 0.05 gives the exact one-sided 95% older bound

$$U(T_{\text{upper}}) = U(A)\,0.05^{-1/n}.$$

The absolute recovery rate cancels. For constant w this is the Strauss & Sadler (1989) and Marshall (1990) expression with a fixed cutoff, which uses all n observations. Marshall (1994, 1997) and Wang et al. (2016) relax the assumption of uniform recovery in other ways. A bound beyond the declared

history's domain (469.4 Ma, where the held PBDB effort series ends) is reported as *not bounded within the domain*, not as an infinite-age inference.

Age envelope. The primary age treatment uses the greatest possible oldest age and the smallest number of units guaranteed older than c. The bound increases with A and decreases with n, so this is an envelope of the exact-age bounds whenever every reported age interval contains the true age. That geological condition is separate from statistical coverage. Independent-uniform and shared-quantile allocations (256 draws, seed 20261102) are reported as sensitivities. Their quantiles are not confidence bounds (Table 8).

Histories. Six histories are fixed without fitting them to the focal fossils:

1. constant recovery in calendar time;
2. PBDB all-taxa land and marginal-marine collections per Myr, apportioned by age overlap;
3. the same, with collections matching the held plant-filtered extract removed;
4. exponential decline into the past, 25 Myr timescale;
5. exponential decline into the past, 10 Myr timescale;
6. the plant-excluded density multiplied by the 10 Myr decline.

Plant exclusion is a response-related filter, not verified independent effort. No proxy establishes actual search opportunities. Cutoffs are the ends of the Emsian, Pragian, Lochkovian and Pridoli.

Shape diagnostics. Conditional on the maximum, the remaining unit ages should be distributed as $U(a)/U(A)$. Kolmogorov–Smirnov statistics at midpoint ages and across age draws expose histories that fit the observed sequence poorly (Table 7). Nominal exact-age *p* values and their spread across draws are not a combined test for interval-dated fossils, a model-selection criterion or an empirical calibration.

S5.3 Results and sensitivities

Table S 3: Conservative 95% older bounds (Ma) by category, reading and recovery history; Emsian cutoff. Units: guaranteed/possible. "None ≤ 469.4": no bound within the declared domain.

Reading	Category	Units	Oldest possible	Constant	PBDB collection density	PBDB density, plant matches removed	Decline into past, 25 Myr	Decline into past, 10 Myr	Plant-excluded PBDB × 10 Myr decline
main	Any land-plant body	75/77	432.9	434.51	436.38	435.73	437.16	none ≤ 469.4	none ≤ 469.4
main	Polysporangiophyte body	70/72	432.9	434.62	436.63	435.94	437.50	none ≤ 469.4	none ≤ 469.4
main	Crown vascular-plant body	48/50	426.7	428.84	428.88	428.45	431.63	none ≤ 469.4	none ≤ 469.4
inclusive	Any land-plant body	79/81	445.2	447.20	448.27	447.85	452.98	none ≤ 469.4	none ≤ 469.4

Reading	Category	Units	Oldest possible	Constant	PBDB collection density	PBDB density, plant matches removed	Decline into past, 25 Myr	Decline into past, 10 Myr	Plant-excluded PBDB × 10 Myr decline
inclusive	Polysporangiophyte body	74/78	445.2	447.34	448.48	448.03	453.62	none ≤ 469.4	none ≤ 469.4
inclusive	Crown vascular-plant body	53/54	426.7	428.63	428.67	428.28	431.10	none ≤ 469.4	468.65

Table S 4: Crown vascular-plant bounds by younger cutoff, main reading.

Younger cutoff (Ma)	Units	Constant	PBDB collection density	PBDB density, plant matches removed	Decline into past, 25 Myr	Decline into past, 10 Myr	Plant-excluded PBDB × 10 Myr decline
Emsian (393.47)	48	428.84	428.88	428.45	431.63	none ≤ 469.4	none ≤ 469.4
Pragian (410.62)	25	428.75	429.71	429.43	429.75	433.80	451.56
Lochkovian (413.02)	18	429.18	430.09	429.81	430.24	434.25	447.52
Pridoli (419.62)	13	428.53	428.54	428.35	428.92	429.80	429.55

Table S 5: Crown vascular-plant bounds under alternative units and source choices (main reading, Emsian cutoff). Region–stage cells use midpoint stage and are a deliberately coarse dependence diagnostic.

B2 unit or source variant	Units	Constant	PBDB density	Decline 25 Myr
Individual collections/observations	78	428.00	428.03	429.58
Locality–age proxies (primary)	48	428.84	428.88	431.63
Whole localities	47	428.89	428.93	431.75
Region–stage cells	20	432.07	431.91	441.60
Wide-age records restored	73	428.09	428.12	429.79
Uncertain material restored	54	428.60	428.63	431.01
Low-confidence names omitted	46	428.94	428.98	431.87
Weak botanical source omitted	48	428.84	428.88	431.63
Kosov/Podolia links collapsed	47	428.89	428.93	431.75
Curated oldest-record set only	15	434.05	434.62	450.50

Table S 6: Leave-one-region-out bounds for crown vascular-plant members, with PBDB exposure recomputed after the same omission. Regions are approximate modern country/state assignments.

Region left out	Units	Oldest possible	Constant	PBDB density
Gondwana Australia Antarctica	44	425.0	427.22	427.33
Gondwana South America	47	426.7	428.89	428.87
Tarim	47	426.7	428.89	428.92
peri Gondwana Europe	47	426.7	428.89	428.85
Baltica	45	426.7	428.99	431.80
Kazakhstania	45	426.7	428.99	429.03
Avalonia	44	426.7	429.04	428.95
South China	43	426.7	429.10	429.06
multiple	42	426.7	429.16	428.74
Laurentia	28	426.7	430.45	429.58

Table S 7: Within-record shape diagnostics, main reading, independent-uniform age allocations.

Category	History	KS D at midpoint ages	Nominal p, 5–95% of age draws
Any land-plant body	Constant	0.194	0.0041–0.062
Any land-plant body	PBDB collection density	0.174	0.0075–0.088
Any land-plant body	PBDB density, plant matches removed	0.214	0.00054–0.039
Any land-plant body	Decline into past, 25 Myr	0.331	6.2e-09–2.7e-06
Any land-plant body	Decline into past, 10 Myr	0.549	2.3e-22–4e-17
Any land-plant body	Plant-excluded PBDB × 10 Myr decline	0.363	2e-10–5.4e-08
Polysporangiophyte body	Constant	0.204	0.0046–0.068
Polysporangiophyte body	PBDB collection density	0.185	0.011–0.13
Polysporangiophyte body	PBDB density, plant matches removed	0.224	0.00086–0.032
Polysporangiophyte body	Decline into past, 25 Myr	0.328	3.1e-08–1.8e-05
Polysporangiophyte body	Decline into past, 10 Myr	0.546	1.9e-20–1.8e-15
Polysporangiophyte body	Plant-excluded PBDB × 10 Myr decline	0.346	1e-08–1.6e-06
Crown vascular-plant body	Constant	0.236	0.073–0.76
Crown vascular-plant body	PBDB collection density	0.234	0.014–0.27
Crown vascular-plant body	PBDB density, plant matches removed	0.320	0.00019–0.016

Category	History	KS D at midpoint ages	Nominal p, 5–95% of age draws
Crown vascular-plant body	Decline into past, 25 Myr	0.363	7e-05–0.016
Crown vascular-plant body	Decline into past, 10 Myr	0.549	2.2e-11–8.4e-07
Crown vascular-plant body	Plant-excluded PBDB × 10 Myr decline	0.347	0.0011–0.044

Table S 8: Age-allocation sensitivity (Emsian cutoff, main reading). Quantiles across allocations are descriptive, not confidence bounds.

Category	History	Allocation	5%	Median	95%	Unbounded fraction
Any land-plant body	Constant	independent uniform	432.25	433.34	434.42	0.00
Any land-plant body	PBDB collection density	independent uniform	432.02	433.42	436.18	0.00
Any land-plant body	Decline into past, 25 Myr	independent uniform	434.48	435.76	437.01	0.00
Any land-plant body	Constant	shared quantile	432.23	433.27	434.35	0.00
Any land-plant body	PBDB collection density	shared quantile	432.00	433.27	435.99	0.00
Any land-plant body	Decline into past, 25 Myr	shared quantile	434.48	435.66	436.90	0.00
Polysporangiophyte body	Constant	independent uniform	432.32	433.33	434.43	0.00
Polysporangiophyte body	PBDB collection density	independent uniform	432.07	433.35	436.15	0.00
Polysporangiophyte body	Decline into past, 25 Myr	independent uniform	434.72	435.93	437.22	0.00
Polysporangiophyte body	Constant	shared quantile	432.34	433.38	434.46	0.00
Polysporangiophyte body	PBDB collection density	shared quantile	432.09	433.50	436.23	0.00
Polysporangiophyte body	Decline into past, 25 Myr	shared quantile	434.78	435.97	437.21	0.00
Crown vascular-plant body	Constant	independent uniform	424.31	426.33	428.46	0.00
Crown vascular-plant body	PBDB collection density	independent uniform	425.14	426.71	428.51	0.00
Crown vascular-plant body	Decline into past, 25 Myr	independent uniform	426.19	428.57	431.05	0.00
Crown vascular-plant body	Constant	shared quantile	421.58	424.10	427.03	0.00
Crown vascular-plant body	PBDB collection density	shared quantile	421.22	424.81	427.18	0.00

Category	History	Allocation	5%	Median	95%	Unbounded fraction
Crown vascular-plant body	Decline into past, 25 Myr	shared quantile	423.10	425.92	429.34	0.00

S5.4 Recovery before the oldest record

After seeing that steep smooth declines fit poorly, we evaluated a complete stage-break grid: flat and PBDB baselines; breaks at 419.62, 426.7 and 432.9 Ma; older multipliers 1, 0.5, 0.1 and 0.01; all categories and both readings (144 cells). The 426.7 Ma break coincides with the oldest possible main B2 record, so no main B2 unit's age support is changed and the within-record shape statistic is unchanged. The grid is a retrospective sensitivity; the 0.05 rule supplies no nominal, simultaneous or selection-adjusted confidence for it. The 432.9 Ma break is a null control when the bound lies younger. No member was selected to match a clock (Table 9).

Table S 9: Retrospective stage-break grid for crown vascular-plant members, main reading (`models/f3/loop3/out/suppression_sensitivity.csv`).

Baseline	Break (Ma)	Older multiplier	Changes age support?	Older bound (Ma)	Midpoint shape p
constant	419.62	1	True	428.84	0.00848
constant	419.62	0.5	True	430.52	0.00424
constant	419.62	0.1	True	444.00	0.00223
constant	419.62	0.01	True	none ≤ 469.4	0.00191
constant	426.7	1	False	428.84	0.00848
constant	426.7	0.5	False	430.98	0.00848
constant	426.7	0.1	False	448.10	0.00848
constant	426.7	0.01	False	none ≤ 469.4	0.00848
constant	432.9	1	False	428.84	0.00848
constant	432.9	0.5	False	428.84	0.00848
constant	432.9	0.1	False	428.84	0.00848
constant	432.9	0.01	False	428.84	0.00848
PBDB density	419.62	1	True	428.88	0.00918
PBDB density	419.62	0.5	True	430.60	0.0186
PBDB density	419.62	0.1	True	451.09	0.00729
PBDB density	419.62	0.01	True	none ≤ 469.4	0.00576
PBDB density	426.7	1	False	428.88	0.00918
PBDB density	426.7	0.5	False	430.98	0.00918
PBDB density	426.7	0.1	False	458.47	0.00918

Baseline	Break (Ma)	Older multiplier	Changes age support?	Older bound (Ma)	Midpoint shape p
PBDB density	426.7	0.01	False	none ≤ 469.4	0.00918
PBDB density	432.9	1	False	428.88	0.00918
PBDB density	432.9	0.5	False	428.88	0.00918
PBDB density	432.9	0.1	False	428.88	0.00918
PBDB density	432.9	0.01	False	428.88	0.00918

S5.5 Biological delay

Fixed delays of 0, 5, 10 and 20 Myr map the recoverability bound to a lineage bound and shift the lineage domain by the same amount. A delay is an assumed mapping, not inferred from this record or transferred from the spore comparison (Table 10).

Table S 10: Fixed-delay mapping, constant recovery, main reading.

Category	Assumed delay (Myr)	Recoverability bound	Lineage bound
Any land-plant body	0	434.51	434.51
Any land-plant body	5	434.51	439.51
Any land-plant body	10	434.51	444.51
Any land-plant body	20	434.51	454.51
Polysporangiophyte body	0	434.62	434.62
Polysporangiophyte body	5	434.62	439.62
Polysporangiophyte body	10	434.62	444.62
Polysporangiophyte body	20	434.62	454.62
Crown vascular-plant body	0	428.84	428.84
Crown vascular-plant body	5	428.84	433.84
Crown vascular-plant body	10	428.84	438.84
Crown vascular-plant body	20	428.84	448.84

S5.6 Simulation checks

Each of 144 cells has 3,000 replicates (seed 20261101) at the reviewed unit counts and recoverability endpoints of 432, 445 and 458.2 Ma. Each correctly specified history is checked with exact and stage-censored dates. The checks then mismatch constant and steeply declining histories, repeat underlying units two or four times, and impose 10 or 20 Myr delays with and without the correct adjustment. Coverage counts no-bound outcomes as covering. Claim frequency and coverage among bounded claims are reported separately, because 95% unconditional coverage does not imply 95% coverage after selecting bounded outcomes. These runs verify the declared procedures and expose their failure modes. They do not qualify the fossils (Table 11; all categories in `simulation_results.csv`).

Table S 11: Simulation results for crown vascular-plant counts.

Scenario	True history	Fitted history	Ages	Duplication	Delay adj.	Endpoint (Ma)	Claim frequency	Coverage (Wilson 95%)
matched	Constant	Constant	exact	1	0.0	432.0	1.000	0.955 (0.947–0.962)
matched	Constant	Constant	stage envelope	1	0.0	432.0	1.000	1.000 (0.998–1.000)
matched	Constant	Constant	exact	1	0.0	445.0	1.000	0.949 (0.941–0.957)
matched	Constant	Constant	stage envelope	1	0.0	445.0	1.000	0.985 (0.980–0.989)
matched	Constant	Constant	exact	1	0.0	458.2	1.000	0.945 (0.936–0.953)
matched	Constant	Constant	stage envelope	1	0.0	458.2	1.000	0.983 (0.977–0.987)
matched	Decline into past, 25 Myr	Decline into past, 25 Myr	exact	1	0.0	432.0	1.000	0.945 (0.936–0.952)
matched	Decline into past, 25 Myr	Decline into past, 25 Myr	stage envelope	1	0.0	432.0	1.000	0.956 (0.948–0.963)
matched	Decline into past, 25 Myr	Decline into past, 25 Myr	exact	1	0.0	445.0	1.000	0.958 (0.951–0.965)
matched	Decline into past, 25 Myr	Decline into past, 25 Myr	stage envelope	1	0.0	445.0	1.000	0.990 (0.985–0.993)
matched	Decline into past, 25 Myr	Decline into past, 25 Myr	exact	1	0.0	458.2	0.212	0.941 (0.932–0.949)
matched	Decline into past, 25 Myr	Decline into past, 25 Myr	stage envelope	1	0.0	458.2	0.070	0.966 (0.959–0.972)
matched	Decline into past, 10 Myr	Decline into past, 10 Myr	exact	1	0.0	432.0	0.139	0.950 (0.942–0.958)
matched	Decline into past, 10 Myr	Decline into past, 10 Myr	stage envelope	1	0.0	432.0	0.078	0.998 (0.996–0.999)

Scenario	True history	Fitted history	Ages	Duplication	Delay adj.	Endpoint (Ma)	Claim frequency	Coverage (Wilson 95%)
matched	Decline into past, 10 Myr	Decline into past, 10 Myr	exact	1	0.0	445.0	0.065	0.949 (0.940–0.956)
matched	Decline into past, 10 Myr	Decline into past, 10 Myr	stage envelope	1	0.0	445.0	0.034	0.966 (0.959–0.972)
matched	Decline into past, 10 Myr	Decline into past, 10 Myr	exact	1	0.0	458.2	0.053	0.950 (0.941–0.957)
matched	Decline into past, 10 Myr	Decline into past, 10 Myr	stage envelope	1	0.0	458.2	0.032	0.968 (0.961–0.974)
matched	PBDB collection density	PBDB collection density	exact	1	0.0	432.0	1.000	0.950 (0.942–0.958)
matched	PBDB collection density	PBDB collection density	stage envelope	1	0.0	432.0	1.000	0.999 (0.998–1.000)
matched	PBDB collection density	PBDB collection density	exact	1	0.0	445.0	1.000	0.957 (0.949–0.963)
matched	PBDB collection density	PBDB collection density	stage envelope	1	0.0	445.0	1.000	0.967 (0.960–0.973)
matched	PBDB collection density	PBDB collection density	exact	1	0.0	458.2	1.000	0.953 (0.945–0.960)
matched	PBDB collection density	PBDB collection density	stage envelope	1	0.0	458.2	1.000	0.999 (0.998–1.000)
matched	PBDB density, plant matches removed	PBDB density, plant matches removed	exact	1	0.0	432.0	1.000	0.954 (0.946–0.961)
matched	PBDB density, plant matches removed	PBDB density, plant matches removed	stage envelope	1	0.0	432.0	1.000	1.000 (0.998–1.000)
matched	PBDB density, plant matches removed	PBDB density, plant matches removed	exact	1	0.0	445.0	1.000	0.948 (0.939–0.955)
matched	PBDB density, plant matches removed	PBDB density, plant matches removed	stage envelope	1	0.0	445.0	1.000	0.983 (0.977–0.987)
matched	PBDB density, plant matches removed	PBDB density, plant matches removed	exact	1	0.0	458.2	1.000	0.952 (0.944–0.959)

Scenario	True history	Fitted history	Ages	Duplication	Delay adj.	Endpoint (Ma)	Claim frequency	Coverage (Wilson 95%)
matched	PBDB density, plant matches removed	PBDB density, plant matches removed	stage envelope	1	0.0	458.2	1.000	0.954 (0.946–0.961)
matched	Plant-excluded PBDB × 10 Myr decline	Plant-excluded PBDB × 10 Myr decline	exact	1	0.0	432.0	0.175	0.951 (0.943–0.958)
matched	Plant-excluded PBDB × 10 Myr decline	Plant-excluded PBDB × 10 Myr decline	stage envelope	1	0.0	432.0	0.119	0.992 (0.988–0.995)
matched	Plant-excluded PBDB × 10 Myr decline	Plant-excluded PBDB × 10 Myr decline	exact	1	0.0	445.0	0.069	0.954 (0.946–0.961)
matched	Plant-excluded PBDB × 10 Myr decline	Plant-excluded PBDB × 10 Myr decline	stage envelope	1	0.0	445.0	0.049	0.951 (0.942–0.958)
matched	Plant-excluded PBDB × 10 Myr decline	Plant-excluded PBDB × 10 Myr decline	exact	1	0.0	458.2	0.045	0.956 (0.948–0.962)
matched	Plant-excluded PBDB × 10 Myr decline	Plant-excluded PBDB × 10 Myr decline	stage envelope	1	0.0	458.2	0.040	0.960 (0.953–0.967)
history mismatch	Constant	Decline into past, 10 Myr	exact	1	0.0	432.0	0.000	1.000 (0.999–1.000)
history mismatch	Constant	Decline into past, 10 Myr	exact	1	0.0	445.0	0.000	1.000 (0.999–1.000)
history mismatch	Constant	Decline into past, 10 Myr	exact	1	0.0	458.2	0.000	1.000 (0.999–1.000)
history mismatch	Decline into past, 10 Myr	Constant	exact	1	0.0	432.0	1.000	0.241 (0.226–0.257)
history mismatch	Decline into past, 10 Myr	Constant	exact	1	0.0	445.0	1.000	0.104 (0.094–0.116)
history mismatch	Decline into past, 10 Myr	Constant	exact	1	0.0	458.2	1.000	0.034 (0.028–0.041)
duplicated horizons	Constant	Constant	exact	2	0.0	445.0	1.000	0.782 (0.767–0.797)
duplicated horizons	Constant	Constant	exact	4	0.0	445.0	1.000	0.529 (0.511–0.546)

Scenario	True history	Fitted history	Ages	Duplication	Delay adj.	Endpoint (Ma)	Claim frequency	Coverage (Wilson 95%)
recoverability delay unadjusted	Constant	Constant	exact	1	0.0	432.0	1.000	0.000 (0.000–0.001)
recoverability delay adjusted	Constant	Constant	exact	1	10.0	432.0	1.000	0.947 (0.938–0.954)
recoverability delay unadjusted	Constant	Constant	exact	1	0.0	432.0	1.000	0.000 (0.000–0.001)
recoverability delay adjusted	Constant	Constant	exact	1	20.0	432.0	1.000	0.952 (0.943–0.959)

S5.7 Figures from the reviewed analysis

S6. From specimens to nodes

The specimen-to-node pilot separates specimen observations, accepted event readings, encounter, trait persistence and node interpretation (`data/pilot/`, `docs/pilot-traits.md`). We reused its source-adjudicated combinations, Hooper et al.'s deposited crown-state outputs and the checked stochastic-map analyses without rerunning them. Aperture (X21) and dispersal unit (X26) were modelled separately, so their marginal supports do not define a joint ancestral phenotype. The X26 monad category combines trilete and inaperturate monads; it is not an automatic absence of cryptospory.

For M2, a proximal trilete aperture is modal at the embryophyte crown across the deposited methods and trees. X26 has a dyad-dominated alternative while X21 stays trilete-dominated, so no joint phenotype probability follows. On the converged extant-tree runs, 77 of 3,382 trilete starts depart along the path to crown Tracheophyta: conditional persistence 0.977 (Monte Carlo SE 0.0026). These trees put crown Embryophyta at 508 Ma, not at Hooper et al.'s 457–450 Ma. Fixed-rate retention scenarios to 452.8 Ma give an extant-tree median of 0.989 (95% interval 0.895–0.995) from 459.1 Ma and 0.894 (0.332–0.956) from 515.5 Ma. These are model-conditional aperture retention scenarios, not observed survival, and they are not multiplied into any coverage.

For M3, the *Cooksonia barrandei* holotype links trilete production and multiple sporangia directly, but its tracheids are unknown. The Irish Homerian material has a conditional sporangial interpretation and no direct demonstration of the anatomy known from younger *C. pertoni*. A total-group or stem occurrence and a crown assignment remain distinct. For M1, cryptospore persistence has not been assessed.

S7. Analyses superseded during review

Two analyses were replaced after independent review. The record is kept so that readers can see what changed and why.

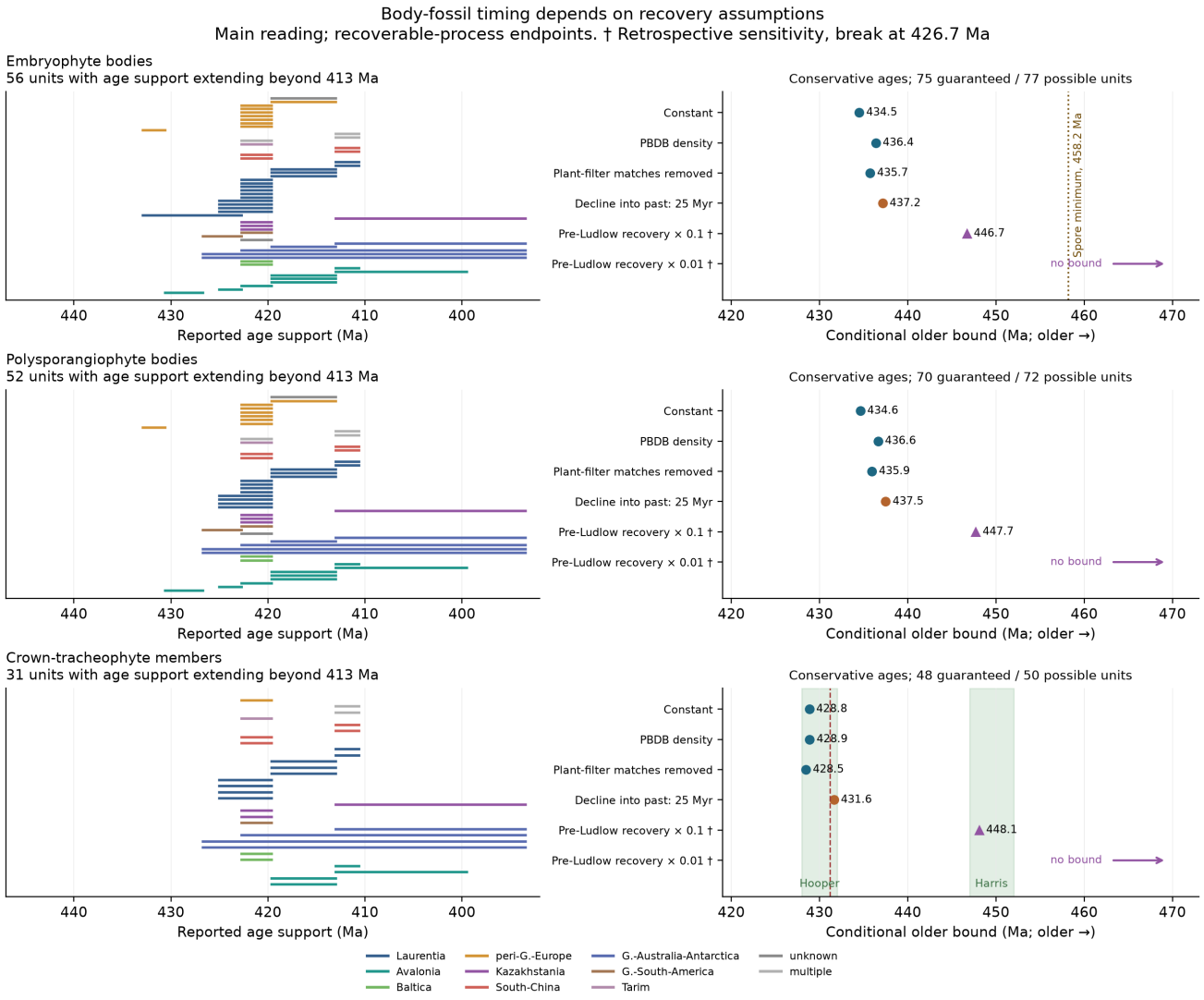


Figure S 3: Reviewed main macrofossil figure as delivered by the analysis (`models/f3/loop3/out/fig_macro_recovery.pdf`).

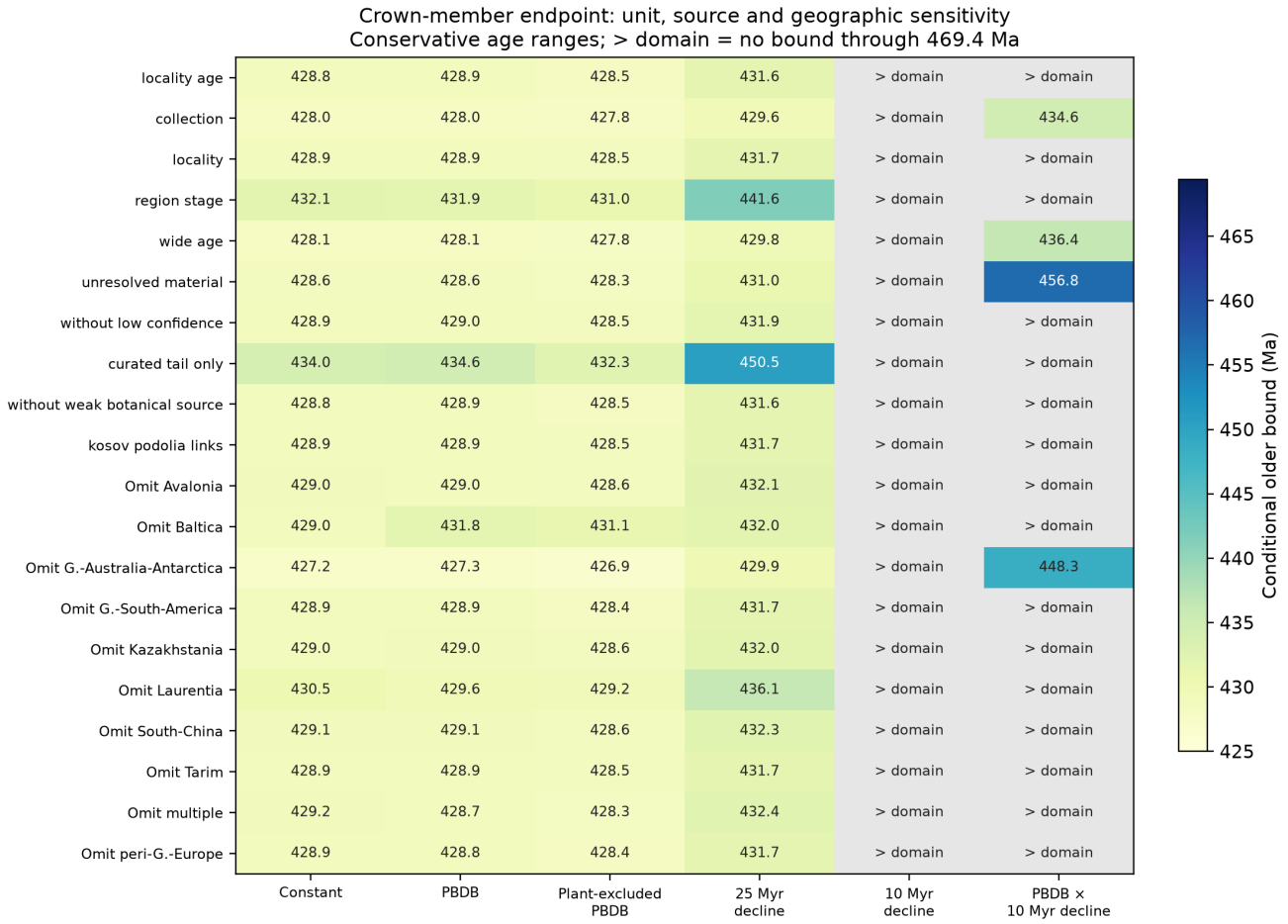


Figure S 4: Sensitivity of the crown vascular-plant bound to units, sources, regions and age rules.

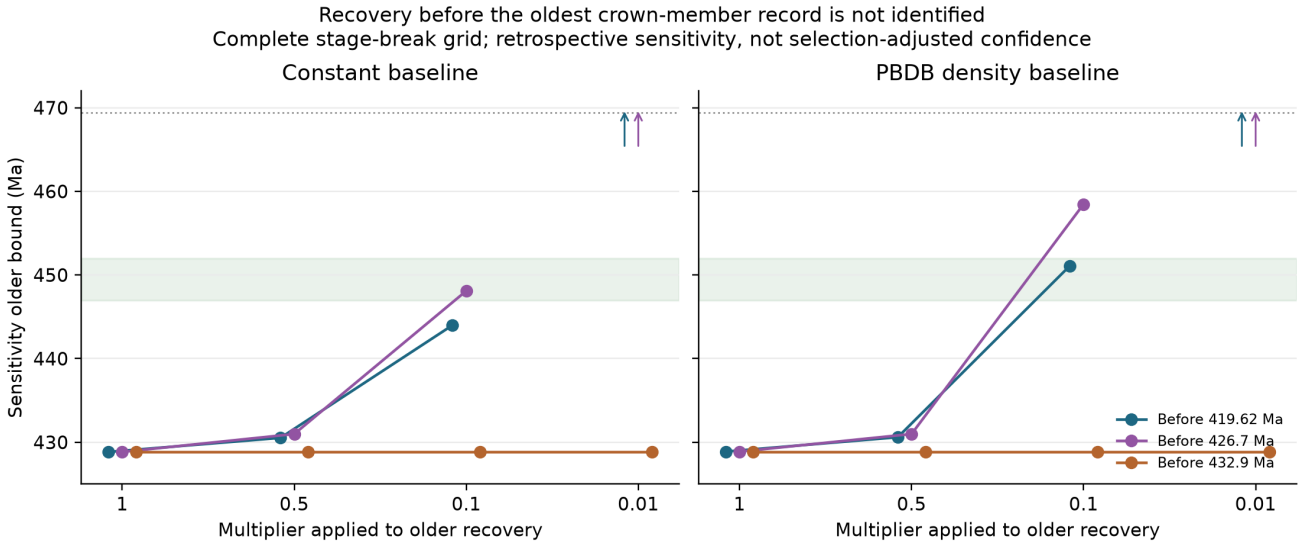


Figure S 5: Retrospective recovery-suppression grid.

The Loop 3 body-record pilot (`docs/loop3-pilot.md`, provisional, by Claude) reported logistic-ramp limits of 452.6, 454.1 and 438.0 Ma for B0, B1 and B2. It read them as favouring the young clock, with a 15–25 Myr preservation lag. Codex's review found:

- stage collection totals passed where per-Myr density was required, counting stage duration twice. The unit correction alone moved B0's logistic limit from 452.57 to 436.78 Ma and flipped the selected capacity between grid boundaries;
- medians of imputed-age limits presented as 95% limits;
- false replacement of distinct PBDB localities through country, genus and age overlap;
- collection numbers treated as independent beds;
- a category gate that let names judged implausible into B1 and B2;
- two different papers sharing the source identifier `wang2026`.

All were repaired. The fixed-history endpoint calculation (S5.2) replaced the profiled ramp. Its numbers and interpretation govern; the pilot's do not (`docs/audits/2026-11-01-loop3-methods-review.md`, `docs/loop3-results.md`).

Loop 2 corrections (`docs/loop2-close.md`, `docs/audits/2026-11-01-loop2-close-claude-response.md`) fixed a misread H0c column, an overgeneralised exponent sensitivity, an unsupported circularity claim and an overstated examination count in an earlier draft.

S8. Reproduction and navigation

From the repository root:

```
uv run --with numpy --with pandas --with scipy --with pytest python -m pytest models/f3/loop3 -q
uv run manuscript/figures.py
python3 manuscript/tables.py
quarto render manuscript/manuscript.qmd
quarto render manuscript/supplement.qmd
python3 manuscript/package.py
```

The body-record analysis is reproduced by `models/f3/loop3/build_reviewed.py`, `analysis.py`, `simulation.py`, `suppression_sensitivity.py` and `figures_reviewed.py`. Hash receipts bind their inputs and outputs. Frozen Loop 1 and Loop 2 runs are retained without regeneration. Third-party deposits are fetched by `data/external/scripts/deposits_fetch.py` and verified against `MANIFEST.tsv`. `manuscript/claims.csv` links each main claim to its evidence and limits.

Supplement references

- Marshall CR (1994). Confidence intervals on stratigraphic ranges: partial relaxation of the assumption of randomly distributed fossil horizons. Paleobiology 20: 459–469. https://doi.org/10.1017/s0094837300012938
- Marshall CR (1997). Confidence intervals on stratigraphic ranges with nonrandom distributions of fossil horizons. Paleobiology 23: 165–173. https://doi.org/10.1017/s0094837300016766
- Wang SC, Everson PJ, Zhou HJ, et al. (2016). Adaptive credible intervals on stratigraphic ranges when recovery potential is unknown. Paleobiology 42: 240–256. https://doi.org/10.1017/pab.2015.37
- Other references as in the main text.